

Lecture 3: Regret Minimisation in Bandits

ID 6002W: Online and Reinforcement Learning

Web-enabled M.Tech. program, IIT Madras

**Where We Are:
Best-Arm Identification in MAB
via Confidence Intervals**

Hoeffding's Inequality

For Confidence Intervals

$$2 \exp(-2n\epsilon^2) \leq \delta \Rightarrow \epsilon \geq \sqrt{\frac{\log(2/\delta)}{2n}}$$

With probability $\geq 1 - \delta$:

$$\mu \in \left[\hat{\mu} - \sqrt{\frac{\log(2/\delta)}{2n}}, \hat{\mu} + \sqrt{\frac{\log(2/\delta)}{2n}} \right]$$

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$n = 10$



$n = 100$



$n = 1000$



95 % confidence intervals

$\delta = 0.05$

Note that confidence intervals are drawn symmetrically around the empirical average

Successive Elimination

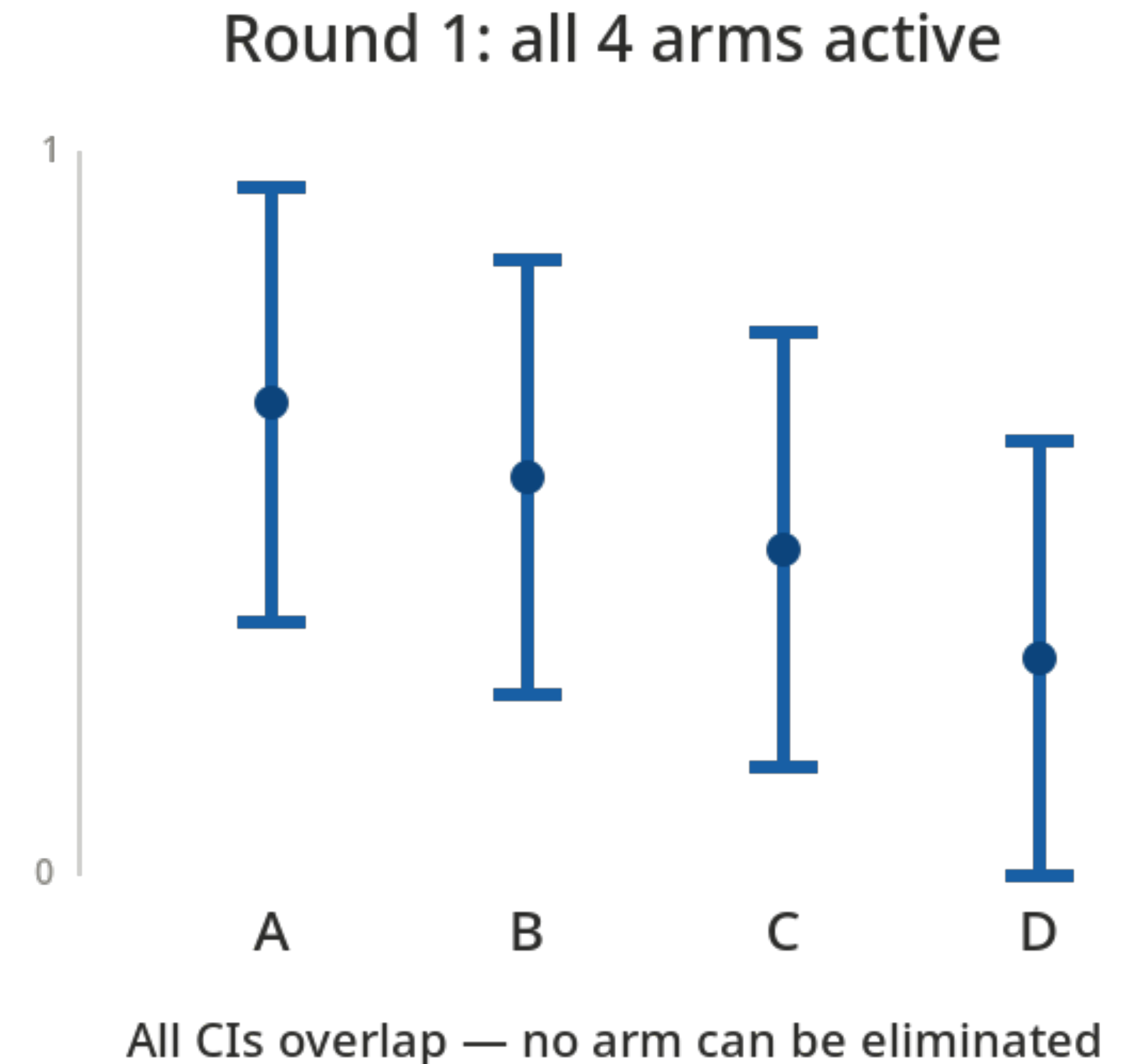
The Key Idea

We have K arms. We want to find the best one with the following strategy.

Maintain a set of **active arms** (initially, all K).

At each round t

- pull every active arm once
- Update *empirical means, confidence intervals*



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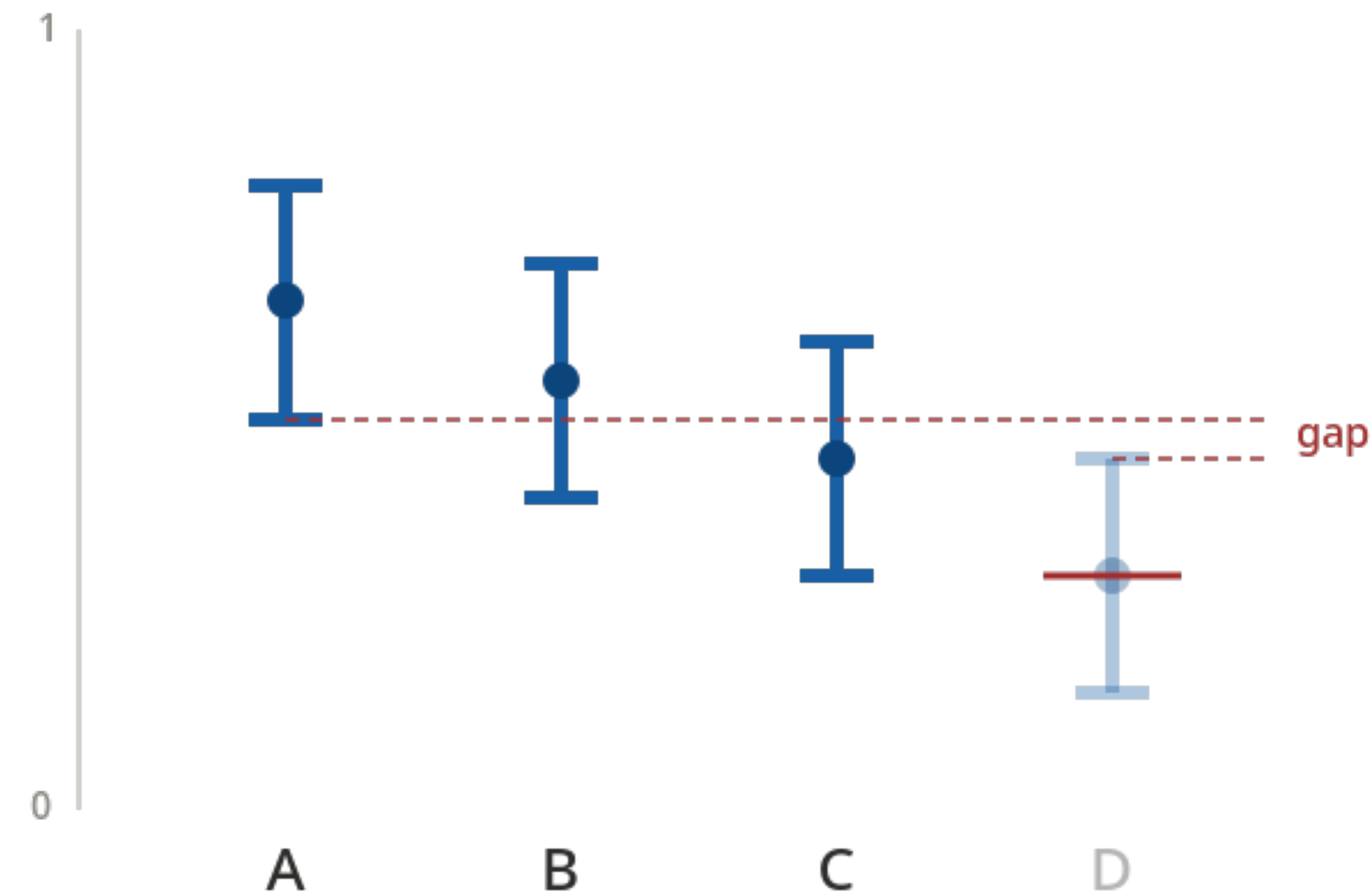
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- pull every active arm once
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- Eliminate any arm whose confidence interval lies *entirely below* another arm's interval

Round 5: D is eliminated



D's CI lies entirely below A's CI

Successive Elimination

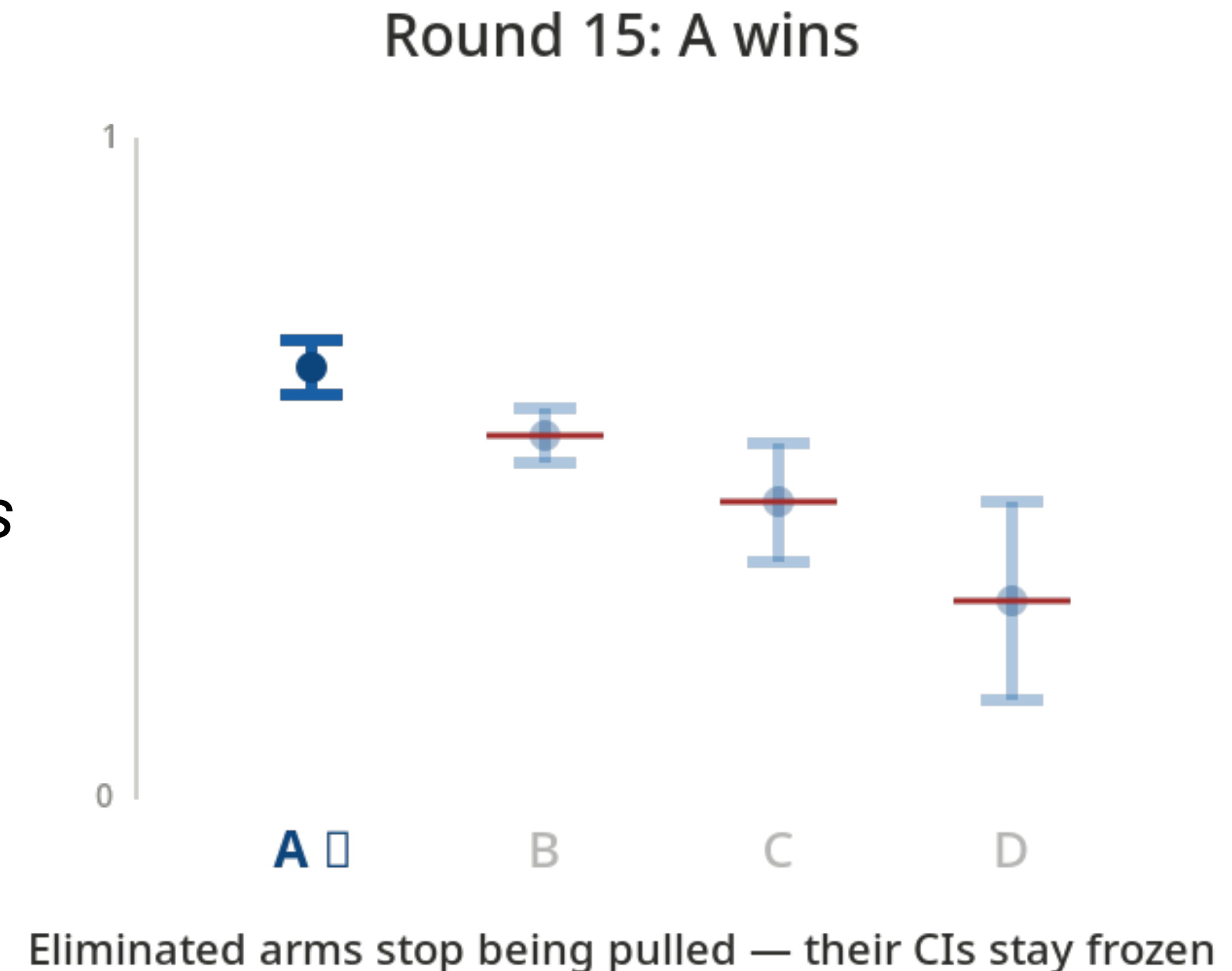
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- Eliminate any arm whose confidence interval lies *entirely below* another arm's interval
- Stop when only one arm remains



**Goal For Today:
A Gentle Introduction to
Regret Minimisation**

Regret

A metric for long-term performance

Recall the definition of regret from Lecture 1:

- If arm distributions were known, then we can always pull arm i^*
 - Get expected reward $T\mu^*$
- Regret = what we missed out on.

$$R^{\text{actual}}(T) = T\mu^* - \sum_{t=1}^T X_t$$

Regret is a random quantity, because both rewards and actions are random

Expected Regret

Cleaner than actual regret

- To simplify analysis, we use expected regret in analysing algorithms:

$$R(T) = \mathbb{E}[R^{\text{actual}}(T)] = T\mu^* - \mathbb{E} \left[\sum_{t=1}^T X_t \right]$$

- Henceforth, we will call this quantity as regret

Expected Regret

Cleaner than actual regret

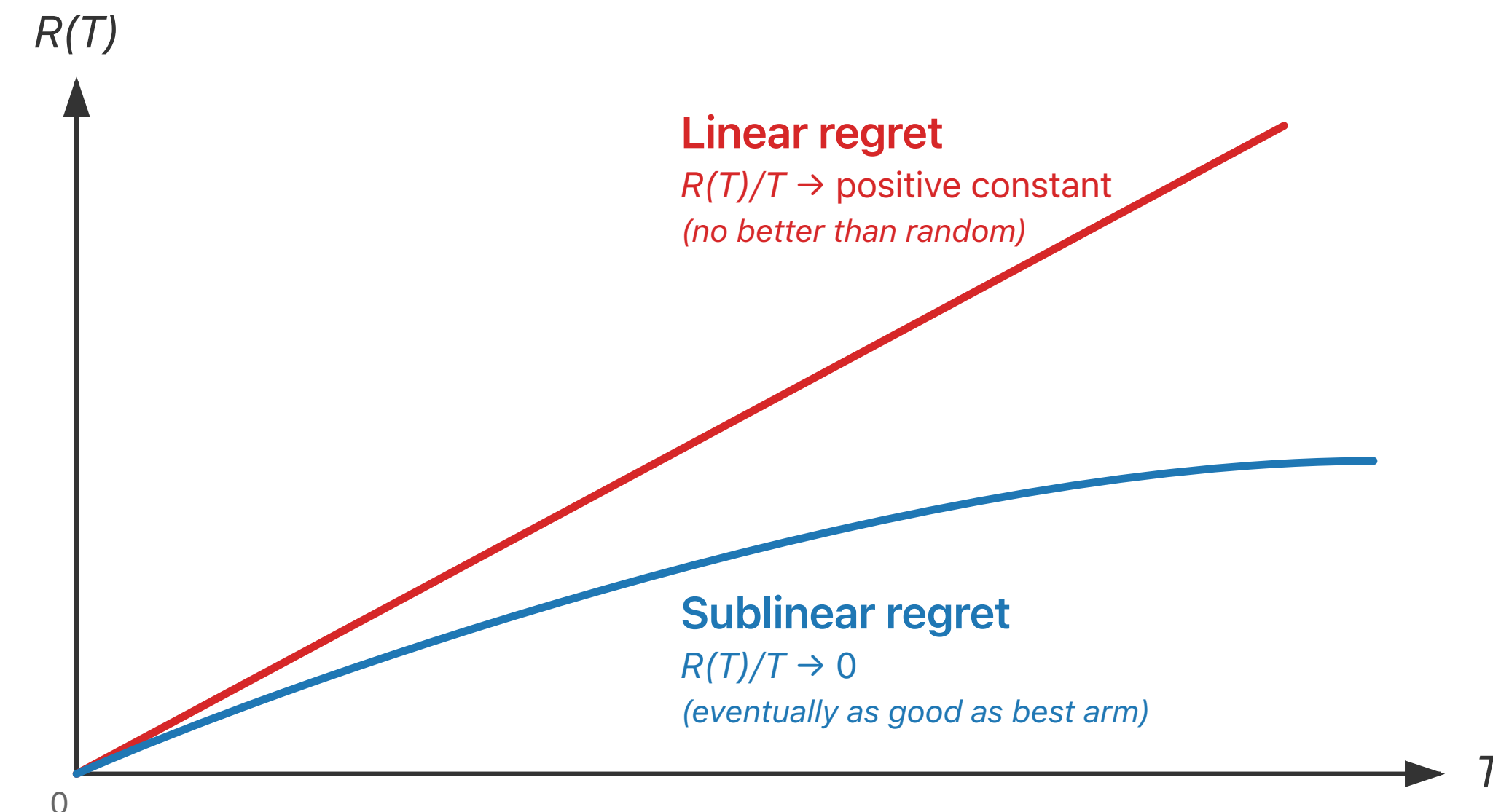
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A good algorithm has sublinear regret!

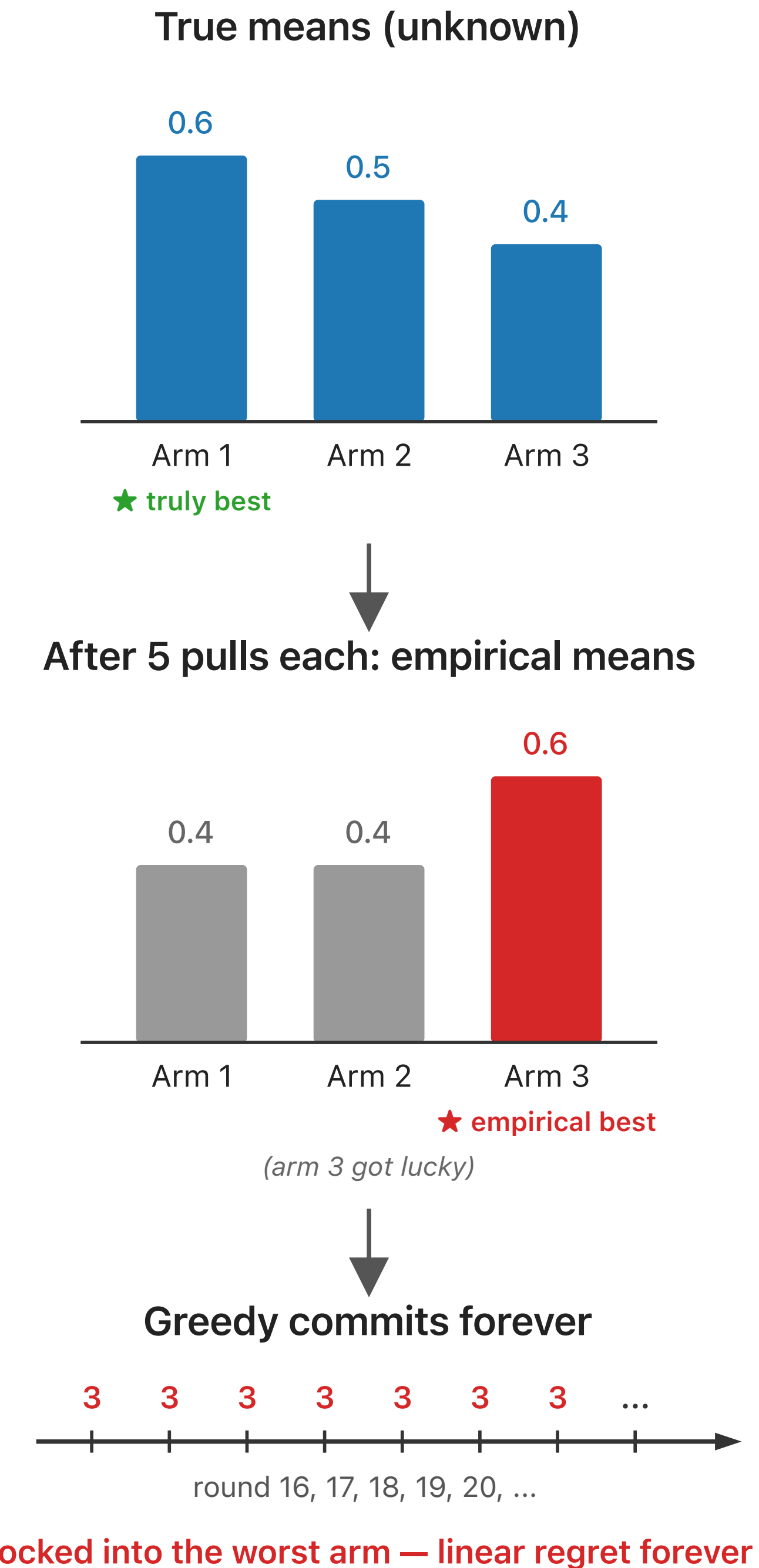
- Equivalently, per-round regret vanishes as T grows
- Algorithm eventually behaves like best arm



Two Naive Strategies

Pure Greedy

- Pull each arm a few times (n)
 - Calculate empirical mean of each arm $\hat{\mu}_i = (\sum_{t:A_t=i} X_t)/n$
- Forever pull the empirically best arm: $\arg \max_{i \in \{1,2,\dots,K\}} \hat{\mu}_i$



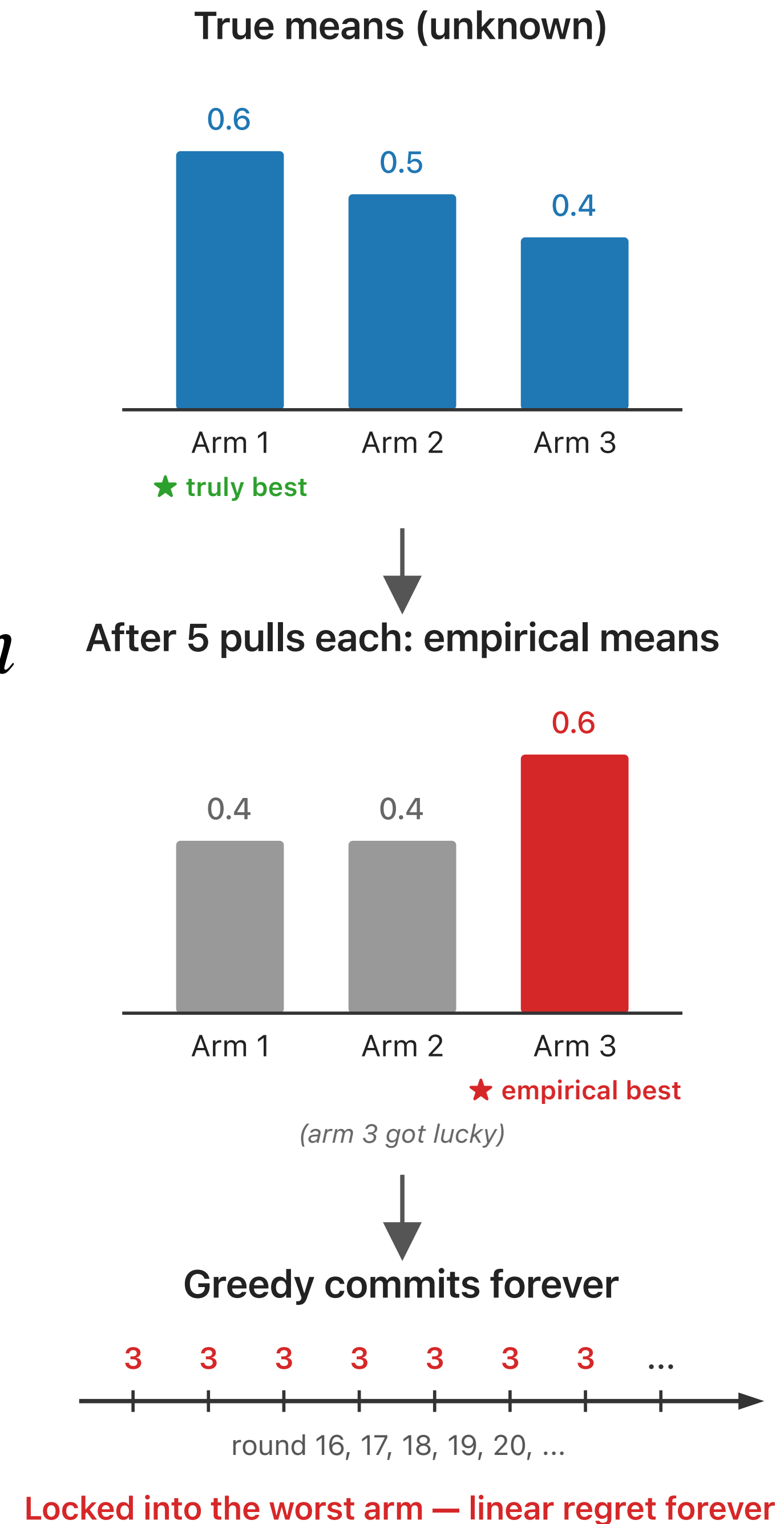
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Pure Random

- Pull a uniformly random arm at every round
 - Equivalently, pull arms in round-robin fashion
- $\hat{\mu}_i \rightarrow \mu_i$, but regret grows linearly in T



ϵ -Greedy Algorithm

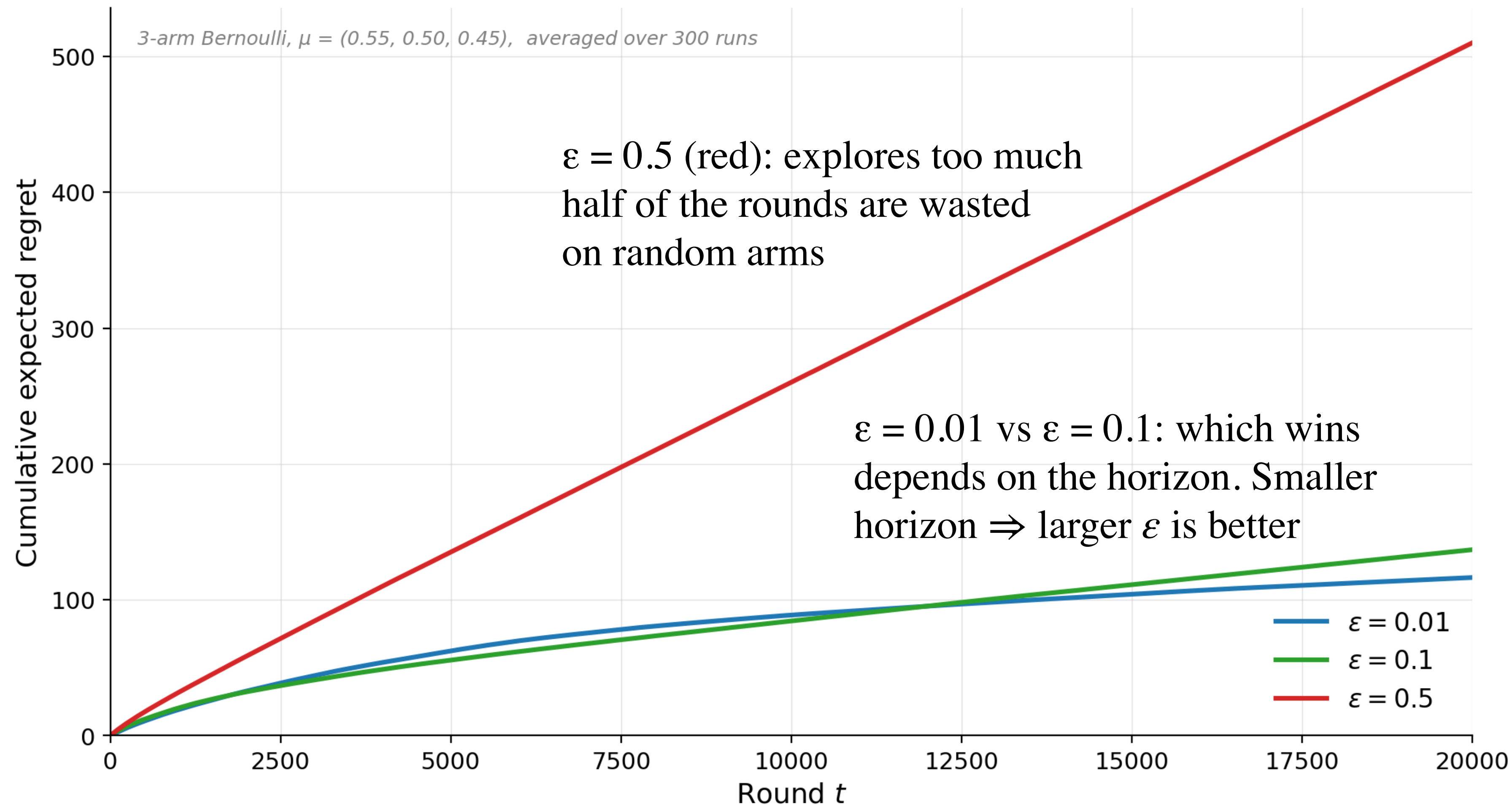
A better solution

- With probability ϵ : pull a uniformly random arm (explore)
- With probability $1 - \epsilon$: pull the empirically best arm (exploit)
- $\epsilon \in (0,1)$ is the exploration rate — a knob you tune

The simplest algorithm that does both exploration and exploitation
by combining pure random and pure greedy strategies

ϵ -Greedy Algorithm

Behaviour and Limitations



All three curves
keep growing —
any fixed ϵ gives
linear regret

No fixed ϵ is universally best

For fixed ϵ , even after we know the best arm, we still explore forever. This leads to linear regret

What if we choose a fixed exploration time, estimate mean rewards, and then commit to best arm?

That is the Explore-Then-Commit algorithm

Where Does Regret Come From?

A useful decomposition

Recall that regret is defined as $R(T) = T\mu^* - \mathbb{E} \left[\sum_{t=1}^T X_t \right]$

- Let A_t denote the action taken in round t
 - Then $\mathbb{E}[X_t \mid A_t] = \mu_{A_t}$

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- Let A_t denote the action taken in round t
 - Then $\mathbb{E}[X_t \mid A_t] = \mu_{A_t}$
- However, A_t is also random!
 - It usually depends on past rewards, which are random

$$\bullet \text{ So } \mathbb{E}[X_t] = \mathbb{E}[\mu_{A_t}] \Rightarrow R(T) = T\mu^* - \mathbb{E} \left[\sum_{t=1}^T X_t \right] = \mathbb{E} \left[\sum_{t=1}^T (\mu^* - \mu_{A_t}) \right]$$

Where Does Regret Come From?

A useful decomposition

Can we simplify the expression $R(T) = \mathbb{E} \left[\sum_{t=1}^T (\mu^* - \mu_{A_t}) \right]$ further?

- Define pull count of arm i : $T_i(T) \Rightarrow \sum_{i=1}^K T_i(T) = T$

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- Define pull count of arm i : $T_i(T) \Rightarrow \sum_{i=1}^K T_i(T) = T$
- If $T_i(T)$ are known for all i ,

$$\sum_{t=1}^T (\mu^* - \mu_{A_t}) = \sum_{i=1}^K \Delta_i T_i(T)$$

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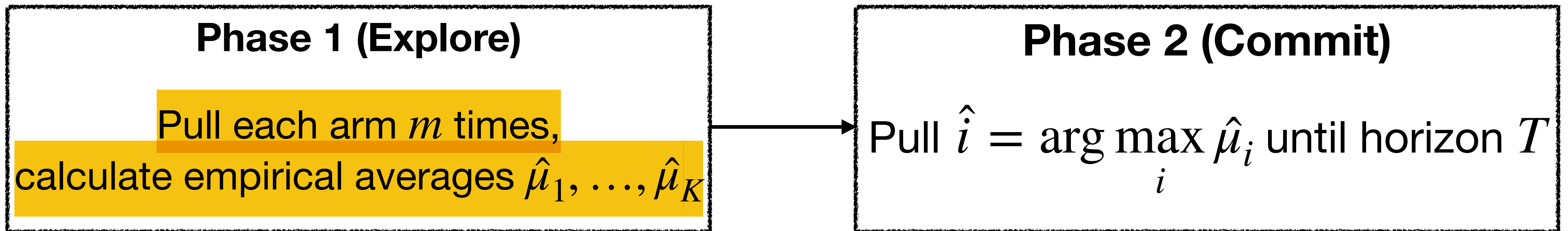
- Taking expectation: $R(T) = \sum_{i=1}^K \Delta_i \mathbb{E}[T_i(T)]$

To minimise regret,
we must minimise
how often we pull
arms with large gaps

Explore-Then-Commit

Basic Idea

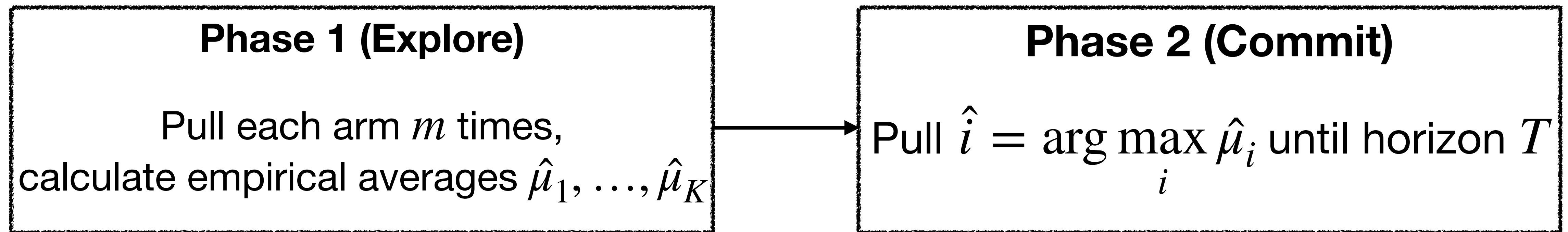
ε -greedy mixes exploration and exploitation at every round.
What if we separate them in time?



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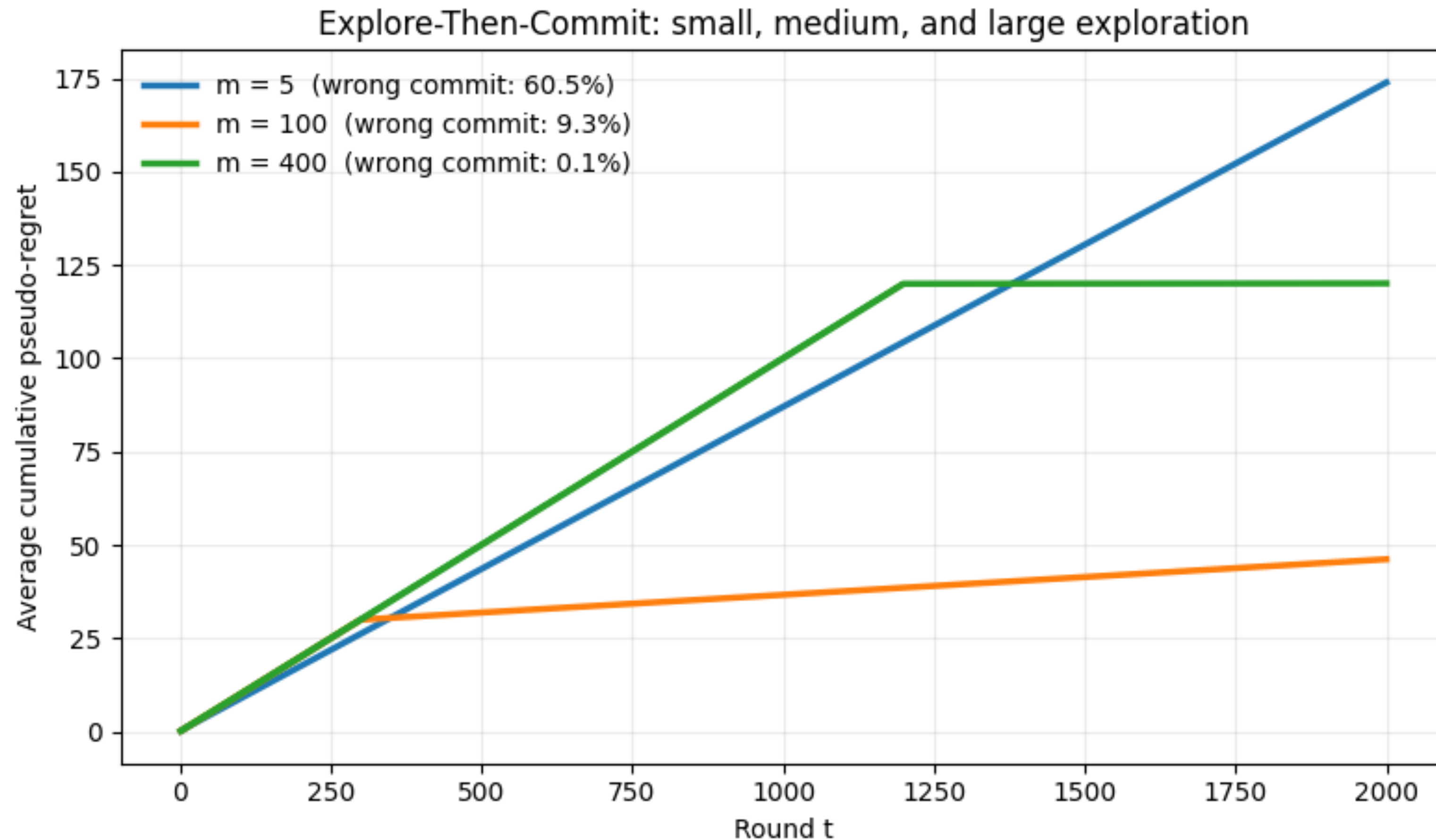


Key parameter: m (exploration budget per arm). Total exploration length: Km

Like Sequential Elimination (SE), it uses samples to identify the best arm.
Unlike SE, stopping time is fixed in advance

Explore-Then-Commit

Visualisation



- When m is small: many runs commit to wrong arm
 $\Rightarrow$ large, linearly growing regret
- When m is large: long exploration phase causes regret to blow up

Medium m balances both costs

Regret comes from two factors: initial exploration (phase 1) + potentially wrong-commit (phase 2)

Explore-Then-Commit

Choosing the right m

- Consider a two-armed bandit with $|\mu_1 - \mu_2| = \Delta$

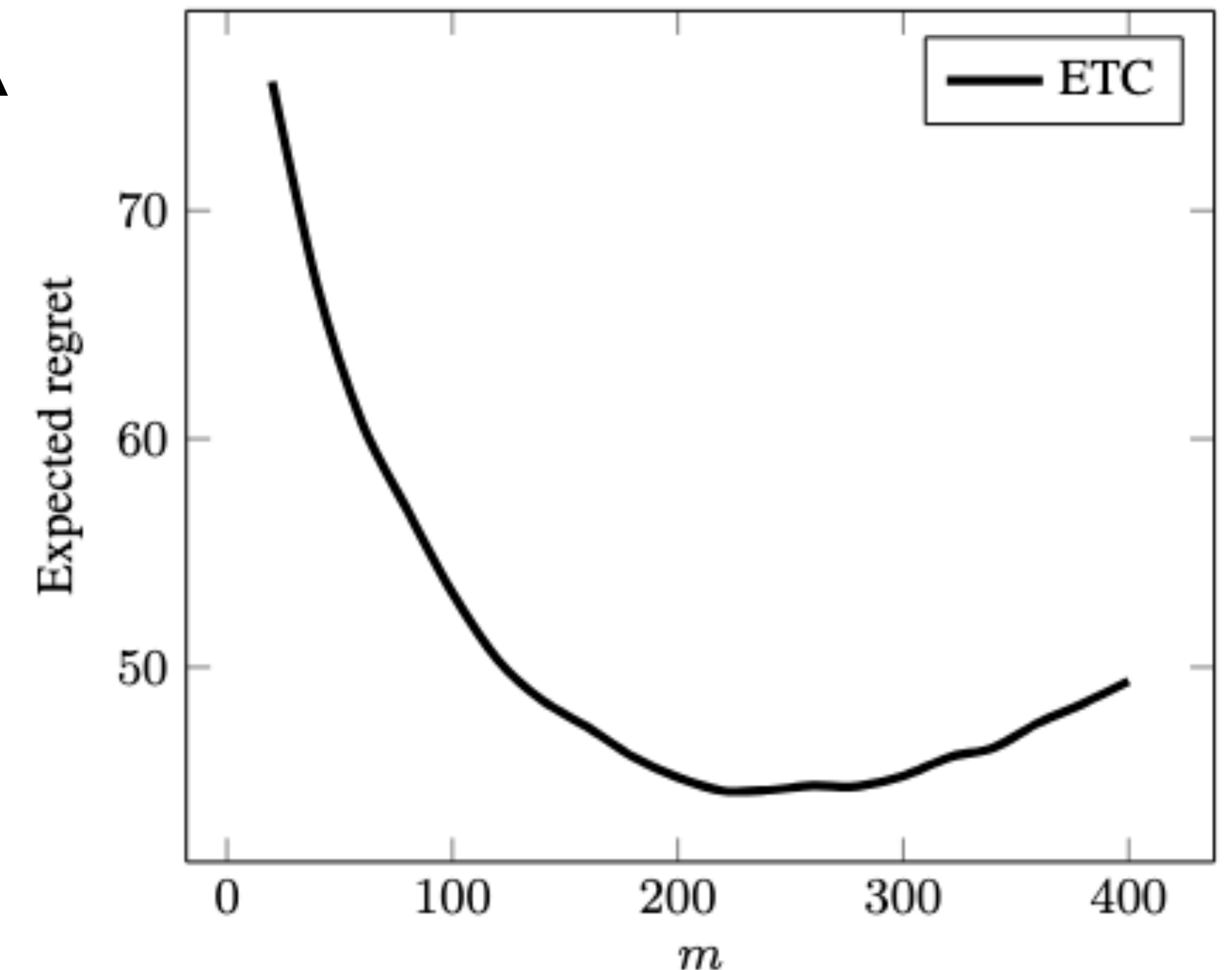


Figure 6.2 from Lattimore & Szepesvari
Figure shows regret as a function of m for
a two-armed Gaussian bandit, $\Delta = 0.1$

Explore-Then-Commit

Choosing the right m

- Consider a two-armed bandit with $|\mu_1 - \mu_2| = \Delta$
- Recall: to distinguish two arms of gap Δ with probability $\geq 1 - \delta$, we need $m \sim \log(1/\delta)/\Delta^2$ samples
- We choose $\delta \propto 1/T \Rightarrow m \sim \frac{\log T}{\Delta^2}$
 - Ensures expected regret from phase 2 is small

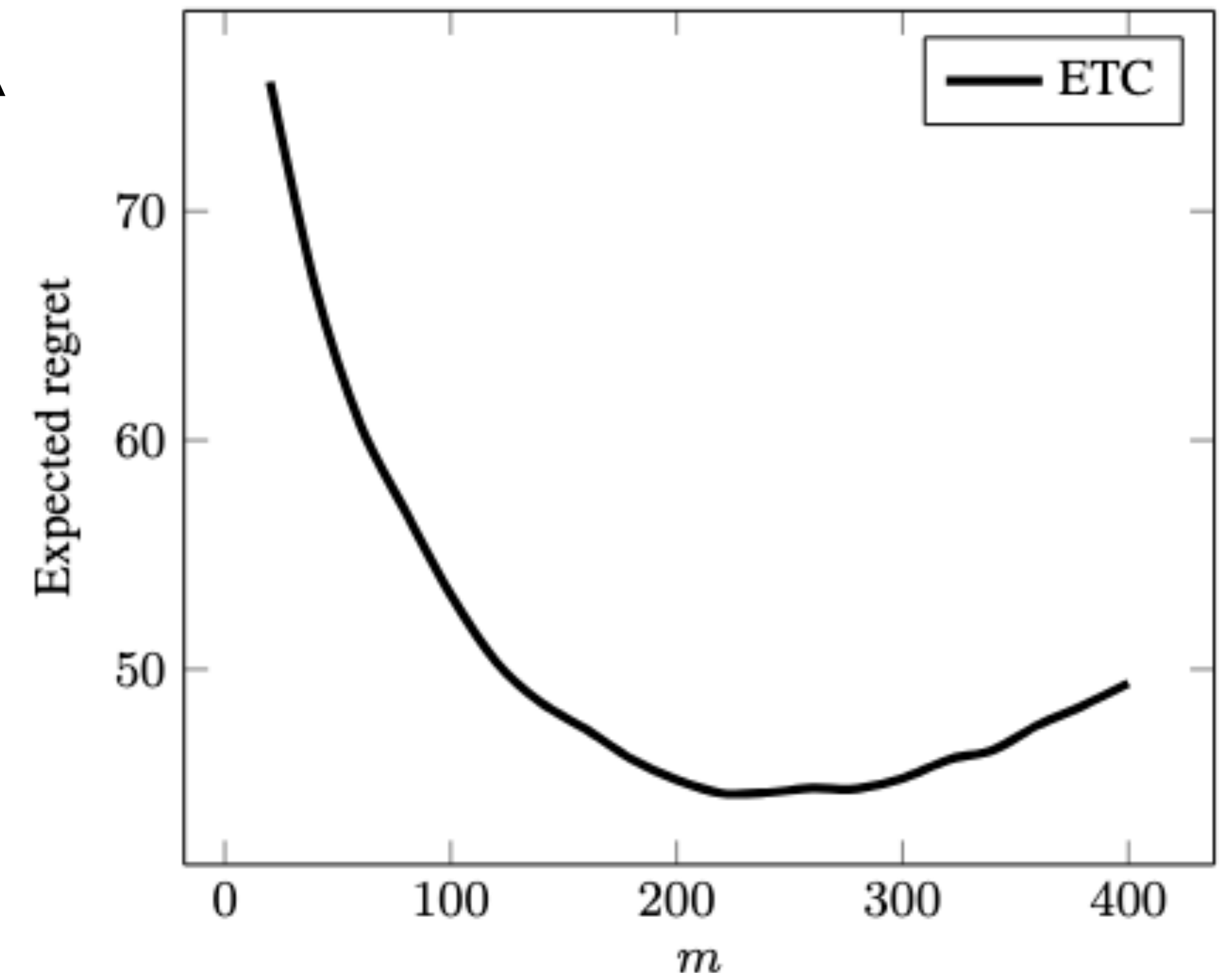


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- Exploration regret $= m\Delta \sim \frac{\log T}{\Delta} \Rightarrow R(T) \sim \log T/\Delta$

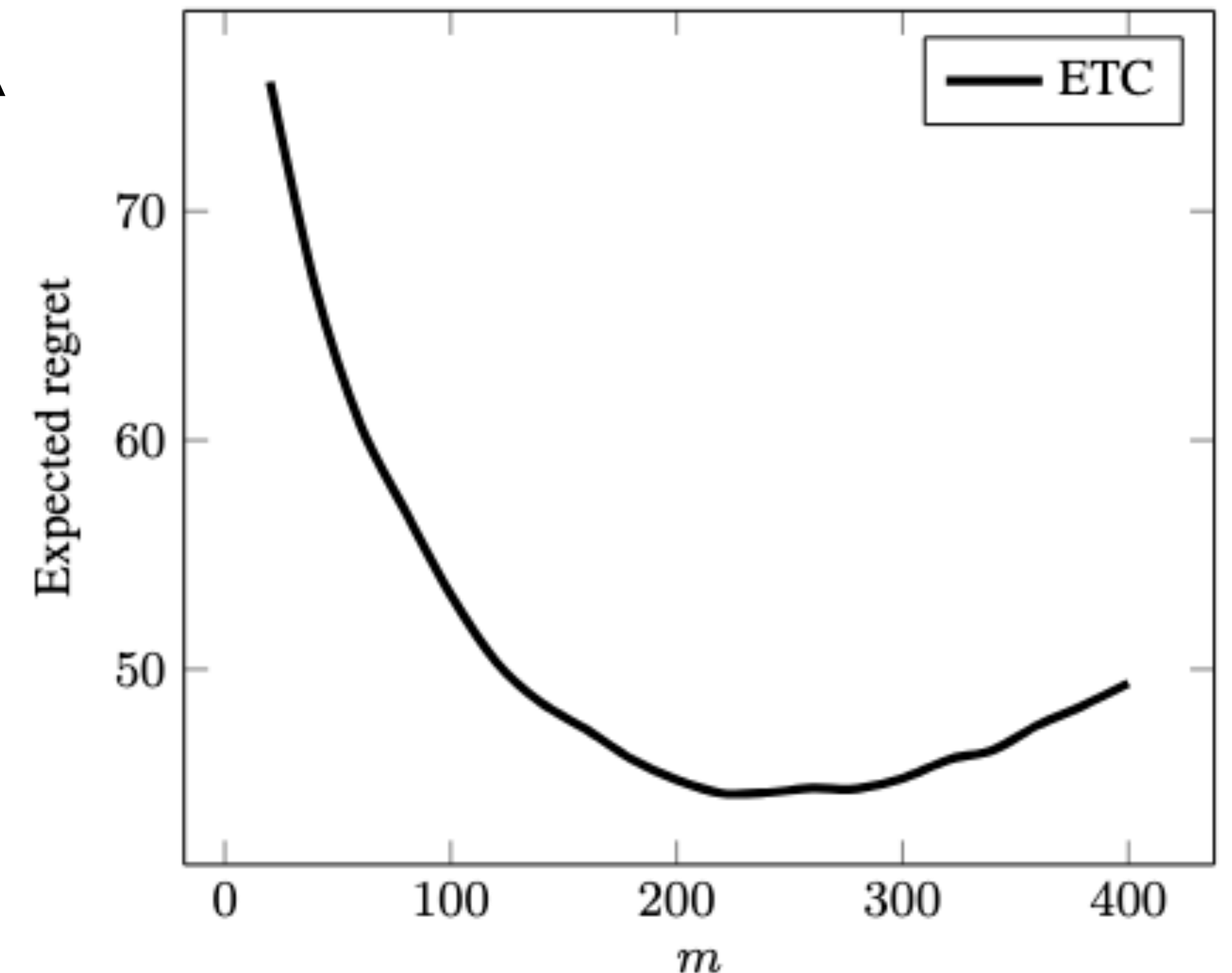


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Explore-Then-Commit

Limitations

If Δ is known:

- Choose $m \sim \log T/\Delta^2$, suffer regret $R(T) \sim \log T/\Delta$

If Δ is unknown:

- Can choose m conservatively to ensure $R(T) \sim T^{2/3}$
- But it does not adapt to easy problems where best arm is clear

Can we avoid choosing m at all?

Need to slowly reduce the extent of exploration over time, using confidence intervals

**The Upper Confidence Bound (UCB) Algorithm
smoothly trades off exploration and exploitation,
giving regret $\log T / \Delta$ without knowing Δ**